

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS58L		Course Title: Database Management Systems Lab				
Total Credits: 1.5 L-0: T-0: P-45		SDGs addressed: 4,9	Course Category: PCCL		Course Type: Laboratory	
Teaching Learning Process		Total Contact Hours/Semester	Assessment Weightage and Marks			
	Lecture	-		C IE	SEE	Total
	Tutorial	-	Weightage, %	40	60	100
	Practical	45	Maximum Marks	40	60	100
	Self-learning and Term Work	-	Minimum Marks	20	25	45*
	Total	45				

Note: *For passing the student must score a Minimum of 45 Marks (CIE+SEE:20 + 25 or 21 + 24)

Course Pre-Requisite: Data Structures.

Course Objectives:

Sl. No.	Course Objectives
1	Learn to design entity relationship and relational models for given user requirements.
2	Formulate SQL statements.
3	Apply database design theory and normalization techniques.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Implement a database schema for a given problem specifications and user-requirements.	L4	WK6
CO2	Use SQL language to create, populate, maintain, and query a database.	L4	WK6, WK8
CO3	Apply normalization theory to validate and revise the logical database design.	L3	WK6, WK8

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

Week	List of Experiments/ Programs	No. of Hours
1	Consider a structure named Student with attributes as SID, NAME, BRANCH, SEMESTER, ADDRESS. Write a program in C/C++/ and perform the following operations using the concept of files. a. Insert a new student b. Modify the address of the student based on SID	03

	c. Delete a student d. List all the students e. List all the students of CSE branch. f. List all the students of CSE branch and reside in Kuvempunagar.	
2	Create a table for the structure Student with attributes as SID, NAME, BRANCH, SEMESTER, ADDRESS, PHONE, EMAIL, Insert atleast 10 tuples and perform the following operations using SQL. a. Insert a new student b. Modify the address of the student based on SID c. Delete a student d. List all the students e. List all the students of CSE branch. f. List all the students of CSE branch and reside in Kuvempunagar.	03
3,4,5,6	Data Definition Language (DDL) commands in RDBMS Consider the database schemas given below. Write ER diagram and schema diagram. The primary keys are underlined and the data types are specified. Create tables for the following schema listed below by properly specifying the primary keys and foreign keys. Enter at least five tuples for each relation. Altering tables, Adding and Dropping different types of constraints. Also adding and dropping fields in to the relational schemas of the listed problems. Delete, Update operations A. Sailors database SAILORS (sid, sname, rating, age) BOAT (bid, bname, color) RSERVERS (sid, bid, date) B. Insurance database PERSON (driver id#: string, name: string, address: string) CAR (regno: string, model: string, year: int) ACCIDENT (report_ number: int, acc_date: date, location: string) OWNS (driver id#: string, regno: string) PARTICIPATED (driver id#:string, regno:string, report_ number: int,damage_amount: int) C. Order processing database Customer (Cust#:int, cname: string, city: string) Order (order#:int, odate: date, cust#: int, order-amt: int) Order-item (order#:int, Item#: int, qty: int) Item (item#:int, unitprice: int) Shipment (order#:int, warehouse#: int, ship-date: date) Warehouse (warehouse#:int, city: string) D. Student enrollment in courses and books adopted for each course STUDENT (regno: string, name: string, major: string, bdate: date) COURSE (course#:int, cname: string, dept: string) ENROLL (regno:string, course#: int,sem: int,marks: int) BOOK-ADOPTION (course#:int, sem: int, book-ISBN: int)	03

	TEXT (book-ISBN: int, book-title: string, publisher: string, author: string) E. Company Database: EMPLOYEE (SSN, Name, Address, Sex, Salary, SuperSSN, DNo) DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate) DLOCATION (DNo,DLoc) PROJECT (PNo, PName, PLocation, DNo) WORKS_ON (SSN, PNo, Hours)	
7,8,9,10	Data Manipulation Language (DML) and Data Control Language (DCL) Write valid DML statements to retrieve tuples from the databases. The query may contain appropriate DML and DCL commands such as: Select with – %like, between, where clause – Order by – Set Operations – Exists and not exists – Join operations – Aggregate functions – Group by – Group by having – Nested and correlated nested Queries Grant and revoke permission	03
11,12	Views and Triggers i. Views: creation and manipulating content. ii. Triggers: creation and execution of database triggers on every insert, delete and update operation.	03
13	Revision	03
14	Revision	03
15	Laboratory Test: Note (Question no. 1 and 2 only for practice)	03

Text Books:

Sl. No.	Author/s	Title	Publisher Details
1	Elmasri and Navathe	Fundamentals of Database Systems	7th Edition, Pearson Education, 2016.
2	Raghu Ramakrishnan and Johannes Gehrke	Database Management Systems	3rd Edition, McGraw-Hill, 2015.

Reference Books:

Sl. No	Author/s	Title	Publisher Details
1	Silberschatz, Korth and Sudharshan	Data base System Concepts	6th Edition, McGraw-Hill, 2016.
2	C.J. Date, A. Kannan, S. Swamynatham	An Introduction to Database Systems	8th Edition, Pearson Education, 2016.

3	Carlos Coronel, Steven Morris and Peter Rob	Database Systems design, Implementation, and Management	9th Edition, Cengage Learning, 2016.
4	Hector Garcia-Molina, Jeffrey D. Ullman and Jennifer Widom	The Database Systems – The Complete Book	Pearson Prentice Hall, 2017.

Web Resources:

Sl. No.	Web Links
1	http://nptel.ac.in/courses/106106093/
2	https://nptel.ac.in/courses/106/104/106104135/

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	-	-	-	-	-	3	3	-	-	3	-	3
CO2	3	3	-	-	3	-	-	3	3	-	-	3	-	3
CO3	3	3	2	-	3	-	-	3	3	-	-	3	-	3

High – 3, Medium – 2, Low – 1